Tangents and normals

1

a

i $y = 2x + 2$

ii $y = -\frac{x}{2} + 2$

b

i $y = 5x + 5$

ii $y = -\frac{x}{5} + 5$

c

i $y = \frac{x}{6} + \frac{1}{6}$

ii $y = -6x + \frac{1}{6}$

d

i $y = 10x + 4$

ii $y = -\frac{x}{10} + 4$

2

a

i $\frac{e}{2}$

ii $ex - 2y + e - \frac{2}{e} = 0$

b

i $-\frac{1}{2e}$

i $x + 2ey - 2e^2 - 1 = 0$

c

i $-\frac{e^2}{e^2 + 1}$

i $e^2x + \left(e^2 + 1\right)y - 4e^2 + \frac{1}{e^2} - 1 = 0$

3 $k = 3$

4

a $y' = (1 - x)\,e^{-x}$

b $\frac{e^3}{2}$

c $y = \frac{e^3}{2}(x - 3) + 3e^{-3}$

5

a $3e$

b $y = 3ex - 2e$

c $A = \left(\frac{2}{3}, 0\right)$

d $B = (0, -2e)$

e $\frac{2e}{3}$ units^2

6

a $(0, 3)$

b -3

c 72°

d -1

e 45°

f 27°

7

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a 81.9° **b** 81.9° **c** 16.2°

8 No solution provided.

Exponential graphs

9

a $f(0) = 2$

b $f'(0) = 1$

c Increasing, because $f'(x) > 0$ for all real x.

d 3

10

a $f(0) = \frac{4}{5}$ **b** $f'(t) = \frac{12}{e^t(2+3e^{-t})^2}$

c Increasing, because $f'(t) > 0$ for all real t **d** 0

e 2

11

a $(2, -e^2)$ **b** Minimum

12

a $f'(x) = -8xe^{-x^2}$

b

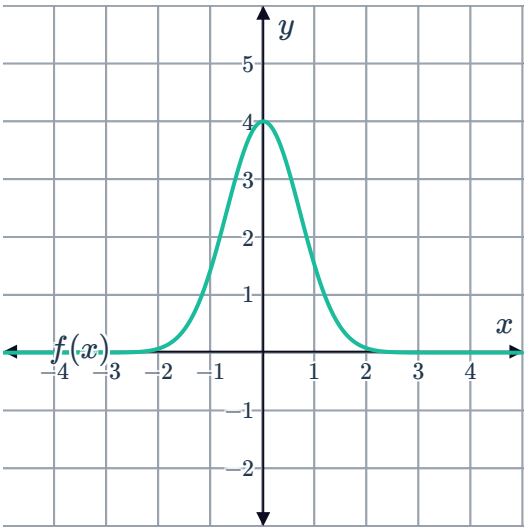
i $x = 0$ **ii** $x < 0$ **iii** $x > 0$

c 0

d 0

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e



13

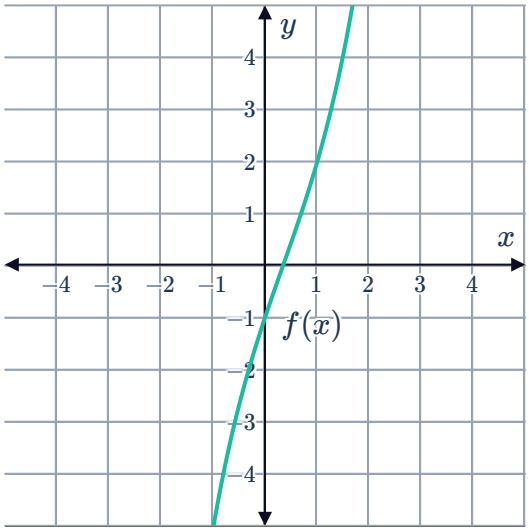
a $x = 0.35$

b $y = -1$

c $y \to \infty$

d $y \to -\infty$

e



14

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a

i $x = 0$

ii $y = 0$

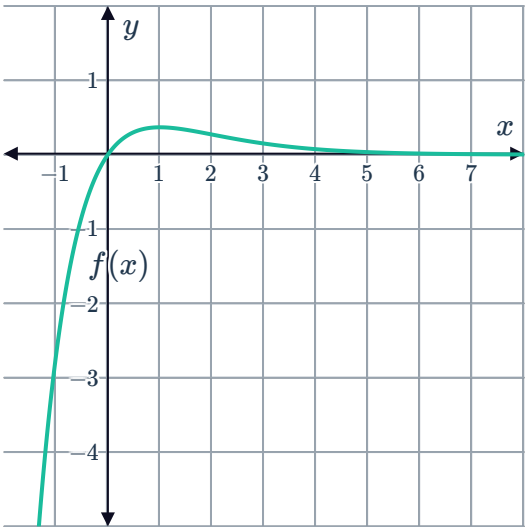
iii $\left(1, \frac{1}{e}\right)$

iv Maximum stationary point

v $\left(2, \frac{2}{e^2}\right)$

vi As $x \to \infty$, $y \to 0$.

vii



b

i $x = \ln 3$

ii $y = -2$

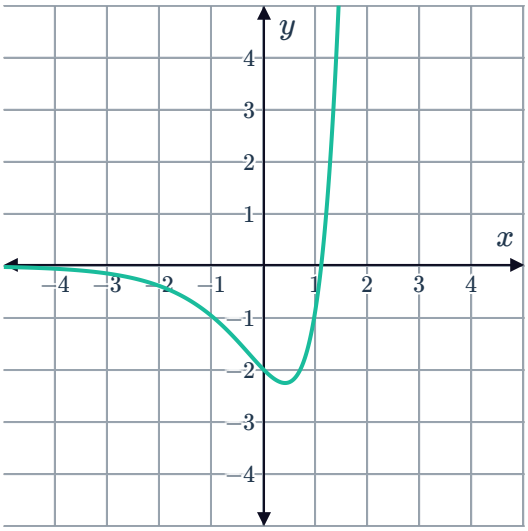
iii $\left(\ln \frac{3}{2}, -\frac{9}{4}\right)$

iv Minimum stationary point

v $\left(\ln \frac{3}{4}, -\frac{27}{16}\right)$

vi As $x \to \infty$, $y \to \infty$.

vii



15

a $x = 0$

b $y = 0$

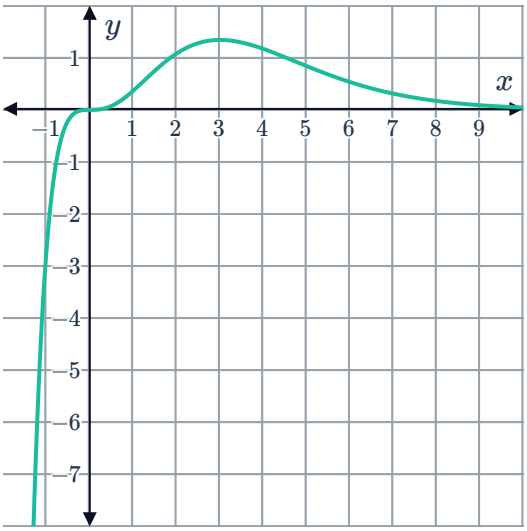
c $y' = 3x^2e^{-x} - x^3e^{-x}$

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d $(0, 0), (3, 1.34)$

e Point of inflection at $x = 0$ and maximum at $x = 3$.

f



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16

a $x = 0$

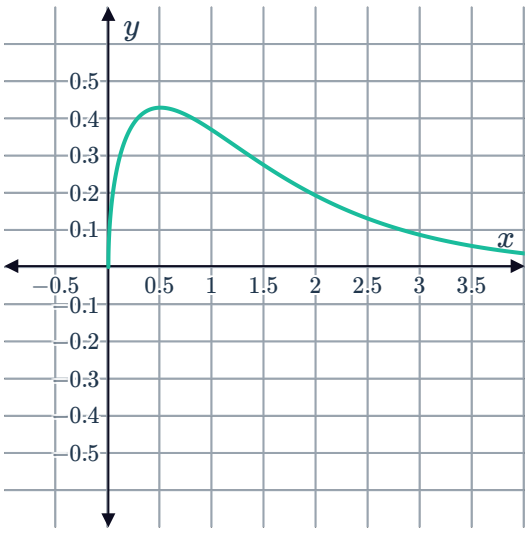
b $y = 0$

c $y' = \dfrac{e^{-x}}{2\sqrt{x}} - e^{-x}\sqrt{x}$

d $(0.50, 0.43)$

e Maximum

f



17

a $x = \ln 3$

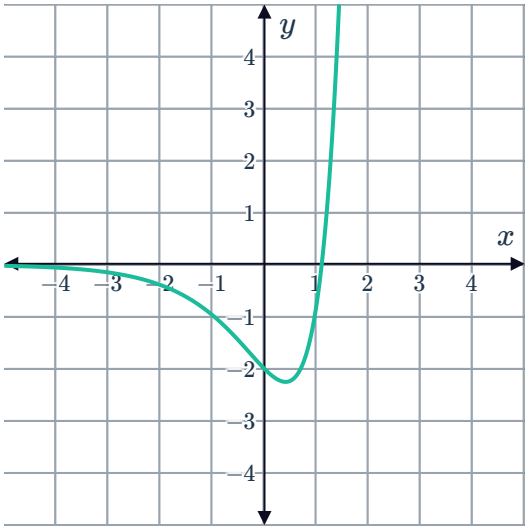
b $y = -2$

c $(0.41, -2.25)$

d Minimum stationary point

e $x \to \infty$, $y \to \infty$
$x \to -\infty$, $y \to 0$

f



Exponential growth and decay

18

a $y = 840\,(0.96)^m$

b 606 rodents

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19

a 8 million

b

i 7130 million **ii** 9286 million

20

a $t = 29$ years **b** $t = 22.3$ years

21

a 1105 km^2 **b** 332 km^2

22 5.08%

23

a 0 **b** $t = \frac{1}{4} \text{ hr}$ **c** 0.736 **d** No

24

a Metal M

b The parameter k.

25

a 1035° C **b** 25° C

26

a $A = -5, B = -170$

b $t = 73$ minutes

c $\frac{dT}{dt} = -0.6$

27

a $f(t) = 1.27\,(0.92)^t$

b 8%

c $\$0.47$

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